

1. Explain the difference between Factory Method and Abstract Factory, in terms of what each one creates(a single product vs. a family of matching products).

Factory Method creates a single type of product, with subclasses deciding the concrete type. Abstract Factory creates a family of related products that are designed to work together, such as buttons and checkboxes with a matching Windows or Mac style.

2. Explain why constructors with many parameters ("telescoping constructors") are hard to use correctly,and describe how the Builder pattern solves this problem.

Telescoping constructors are hard to use because users must remember the meaning and order of many parameters, making mistakes easy. The Builder pattern solves this by using named methods to set parameters step by step, allowing optional parameters to be omitted and making object creation clearer and less error-prone.